

Acute Lymphoblastic Leukemia Prediction from Microscopic Blood Smear Images Using Advanced Machine Learning

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Abstract— Overproduction of the immature white blood cells in the bone marrow leads to leukemia which is a dangerous disease. The probability of recovery can be increased if leukemia is predicted at an early stage. Typically, morphological analysis for the detection of leukemia is performed on blood cells manually by the experienced medical staff which results in various drawbacks like lack of medical staff, slow analysis, and prediction is dependent on the medical staff skills. Therefore, we proposed an automated system which can detect leukemia correctly and at low cost. The proposed methodology is based on the prediction of acute lymphoblastic leukemia by using microscopic blood smear images. For the detection of 'ALL', we have used three baseline architectures i.e. VGG-16, Inception-V3 and ResNet50 that are pre-trained on the ImageNet database. Data augmentation and pre-processing have been utilized to minimize the error of the network. Transfer learning is becoming a developing health image processing method because of its greater presentation in minor samples of images, unlike old-style DL methods. In this study, we have recommended a novel transfer-learning-based automatic ALL classification approach. Our proposed ResNet50 model achieved 100% accuracy on ALL_IDB dataset and achieved 97% accuracy on C_NMC-2019 dataset.

Keywords— Blood Cancer, Medical Imaging, ResNet50, transfer learning, image processing.

I. INTRODUCTION

Cancer is a dangerous disease which occurs due to overproduction of the abnormal cells. These abnormal cells gather around the normal cells and result in inefficient working of the body. There are many types of cancers which are classified on the basis of the type of cell in which they originate [1]. Blood cancer is the most common type of cancer. Functionality and production of the blood cells are affected due to blood cancer. Major types of blood cancer are Leukemia, Myeloma and Lymphoma[2].

Blood-forming cells are found in the bone marrow and leukemia starts from these blood-forming cells. Abnormal white blood cells (WBC) rapidly produced in leukemia. Higher number of abnormal WBC reduces the functionality of bone marrow [2]. Soft fatty tissue inside bone cavities are called bone marrow. Bone marrow contains hematopoietic, fat cells, blood

vessels, fibrous tissue and fluid. Hematopoietic cells are also called stem cells. The development of the blood stem cells further results in the creation of either myeloid stem cells or lymphoid stem cells. Lymphoid stem cells produce lymphocytes which is the type of WBC. On the other hand, myeloid stem cells produce platelets, granulocytes, red blood cells, and monocytes. Monocytes and granulocytes are also types of WBC. The immature generation of WBC by stem cells leads to leukemia. Only one immature or blast cell produces billions of other blast cells [3].

Acute Lymphoblastic Leukemia (ALL) is a cancer that begins from the early class of WBC called lymphocytes in the bone marrow. These affected cells spread into different body parts including liver, lymph nodes, and central nervous system. If it is not cured early, it would probably be incurable within a few months. It is most occurs in children rather than adults [3]. AML starts in the bone marrow but it quickly moves into the blood. It is more common in older people than younger [4]. CLL starts from lymphocytes which is a type of WBC and it is most common leukemia in adults. Most people have no signs for a few years because of the slow development of CLL [5]. Genetic change occurs in myeloid cells which make an abnormal gene that is called BCR-ABL, which turns the cell into a CML cell. CML leukemia cells grow and divide slowly but it can change into AML in which abnormal cells are growing quickly. CML mostly occurs in adults but it rarely affects children [4]. Microscopic Blood smear image (MBSI) of a healthy person and Microscopic Blood smear image (MBSI) of an ALL patient is shown in "Fig. 1", and "Fig. 2", respectively.

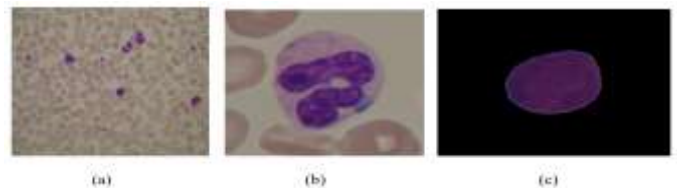


Fig. 1. (a-b) belong to ALL_IDB dataset and (c) belong to C-NMC dataset.

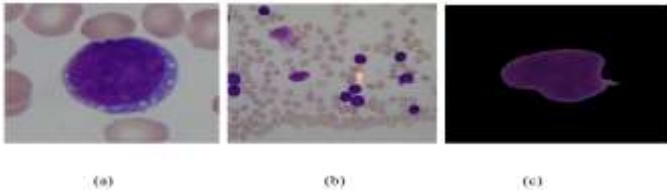


Fig. 2. (a-b) belong to ALL_IDB dataset and (c) belong to C-NMC dataset

The main scope of the research work is the identification of the Acute Lymphoblastic Leukemia by performing different steps. Firstly we will use two ALL-IDB and C-NMC datasets that are available publically. Secondly We will be including preprocessing and data augmentation techniques. Thirdly, Classification will be performed using different pre-trained models based on transfer learning and fine-tuning. Advance machine learning and transfer learning techniques are also used in various medical imaging[6], engineering tasks[7] and versatile applications[8].

This research is useful to increase productivity worldwide and at a similar period can benefit and be a vast influence in the health field. The key purpose is to improve algorithms that can bring data from human blood where human blood is the key basis to distinguish infection at earlier class and can head off it rapidly.

II. LITERATURE REVIEW

Literature reviews included of past studies of traditional machine learning and advanced machine learning techniques.

A. Traditional Machine Learning Methods

In a research Nasir and Mustafa proposed a system for blood cells count using image processing technique. They used 91 images of blood cell images among which 34 images were normal and 57 images were diseased images (14 images for ALL and 43 images for AML). They performed few pre-processing and post-processing techniques. They converted the original image into a grayscale image. Grayscale image was used to make histogram, that histogram divided into three separate regions WBC, red blood cells (RBC) and background. Threshold values were used to segment regions. Using a thresholding grayscale image was converted into a binary image. Binary images had some noise that needed to be removed so image enhancement techniques were used. Two methods were used for image enhancement in this research (i). Removing pixel technique (ii). Gaussian Filter. Ratio of WBC to RBC was calculated through enhanced images.

Amir and others. [9] divide the input image into 2D images that associated with R, G, and B channels. Firstly, they used a feedforward neural network for segmentation, and then used Otsu thresholding algorithm for final segmentation of leukocyte nuclei. Bodzas et al. [10] presented image processing and histogram-based thresholding segmentation techniques to detect WBC nuclei from the MBSI. They used SVM and ANN machine learning algorithms for the classification of healthy and blast cells from leukocyte nuclei.

Raghav et al. [11] remove the noise using multiple filters for getting the preprocessed image. They applied k-means, fuzzy C means and moving k-means for segmentation. Shape based features were extracted using k-mean for the classification of healthy and unhealthy cells.

B. Advance Machine Learning Methods

Payam et al. [12] presented different pre-trained deep learning models for classification of leukemia. They also proposed an ensemble approach combining two architectures, such as NASNetLarge and VGG19 that performed better than individual architectures. Loey et al. [13] adopted two models to detect normal and blast cells from the Leukemia database. First model was based on machine learning algorithms which achieved overall 99.79% accuracy. Second was alexnet pretrained-model which achieved overall 100% accuracy in which used fine-tuning of transfer learning. They used 2820 augmented images for that purpose.

Zakir Ullah et al. [14] introduced CNN based VGG-16 model that used the Efficient channel attention module to extract the high-level features from the image database. They applied image processing and data augmentation techniques on the C-NMC dataset to overcome the overfitting problem. The architecture achieved 91.1% accuracy.

Jiang et al. [15] presented the ViT-CNN ensemble network in which used the vision of CNN based EfficientNet and vision transformer network using Fine-tuning. Before classification they applied preprocessing techniques to raise the condition of the image and increase the data using different augmentation techniques. ViT-CNN network achieved higher accuracy than individually used EfficientNet and vision transformer networks.

III. METHODOLOGY

MBSI images were used in this research to predict ALL and normal cells. This research is helpful for hematologists for identification of ALL and normal cells. Flow of the proposed methodology is briefly described in "Fig. 3".

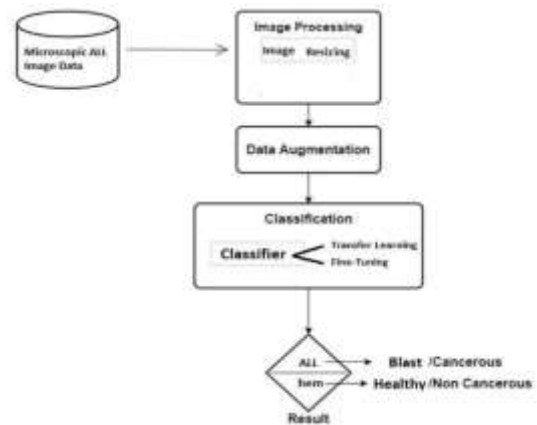


Fig. 3. Flow of Proposed Methodology.

A. Dataset Description

For research purposes, two datasets were used. First dataset was collected from ALL-IDB which is publicly available [16]. In this study, microscopic images of blood samples from Acute Lymphoblastic Leukemia (ALL) Image Database namely ALL-IDB were used for the diagnosis of Leukemia. The images of the dataset have been captured with an optical laboratory microscope coupled with a Canon PowerShot G5 camera. The ALL-IDB images has two dissimilar versions (ALL-IDB1 and ALL-IDB2), which can be freely used from in the dataset each image is classified by expert oncologists. This website provided an application form to get the dataset. Request for the ALL-IDB dataset directed by an email and both datasets were used in this research. Dataset of each class show in Table 1.

TABLE 1. DISTRIBUTION OF ALL_IDB DATASET

Class	No. of images
Normal	189
Blast cells	179
Total	368

Second dataset is collected from microscopic blood smear images, named C-NMC_2019 which was released in the heath imaging challenge, prepared by ISBI 2019 that is publically available[17]. This dataset included blast and healthy cells which consist of a large number of images. Dataset of each class is show in Table 2.

TABLE 2. DISTRIBUTION OF C_NMC DATASET

Distribution of Dataset	Class	No. of the images
Training Dataset	Normal	3,050
	Blast cells	6,544
Validation Data	Normal	170
	Blast Cells	363
Test Data	Normal	170
	Blast cells	364
	Total	10,661

B. Preprocessing

The first phase of proposed methodology is preprocessing of the MBSI image. Preprocessing is the important phase because after this phase we used that image for the next proceeding phase. we converted the size of each image into 224 x 224 resolution in “Fig. 4”.

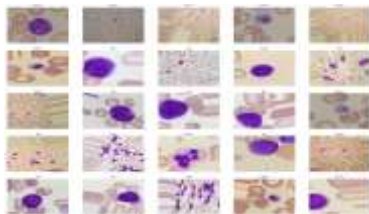


Fig. 4. Augmented some healthy and blast images of ALL_IDB dataset

C. Advance Machine Learning Methods

Augmentation is the second phase of our proposed research which is the most important phase because it affects the overall system accuracy. In most research papers, authors used

different augmentation techniques to solve the problem of less data in the dataset such as flips (vertical and horizontal), cropping the image, translation, image rotating, and so many [11], [13], [18]. From this technique we can increase our training data 10 to 15 times or even more.

ALL_IDB Dataset is not big which we are using in research, so we are needed to artificially enlarge our dataset. We applied rotation, flip(horizontal), zoom and vertical shifting and horizontal shifting on each image in the dataset randomly. These are all the augmented variations that used to solve the problem and increase the number of data to help the training network see Table 3.

Second dataset is imbalanced which can be seen in Table 4. clearly. The contrast dataset can be a trouble in training the network, as hardly any properties are learned from classes with relatively few amounts of data. Two methods used to control the problem of imbalance data in past research, named are oversampling and downsampling. Normally, an oversampling indicates a duplication of the same data randomly [19]. We also applied horizontal flip to create variation on the training dataset which is an augmentation technique.

TABLE 3. DISTRIBUTION OF C_NMC DATASET

Class	No. of Images
Normal	1500
Blast cells	1500
Total	3000

TABLE 4. DISTRIBUTION OF C_NMC DATASET

Class	No. of Images
Normal	3050
Blast Cells	3050
Total	6100

D. Classification

This phase is very important because it is the main building block of the research. In this phase, classify the images into healthy and blast cells. For classification, three types of advanced machine learning pre-trained classifiers are used in proposed research which are VGG16 [20], Inception V3 [21] and Resnet50 [22] to get the best result. For results we will use testing data that will not be used in training. For classification we will apply transfer learning [23] and fine-tuning approaches on pre-trained models [24].

a) Transfer Learning

Transfer learning is used in machine learning that transfers the knowledge to train new data by a previously trained model. This technique consumes less time to train the model and can be effective and useful when the dataset is small to train, because the model weights have already been learned on a very big dataset [25]. We applied transfer learning techniques in such a way, we freezed all the layers including fully connected layers (FC) that were responsible for classification. We added

some new layers on the top of the model using Keras API. Firstly, we added batch normalization layer (BN) followed by add dense fully connected layer (256 or 128 elements) with relu activation function and add dropout layer with 45 percent elements and add another dense layer with 2 outputs and softmax as activation function.

b) Fine-Tuning

Fine-tuning is a kind of transfer learning approach that is applied on the pre-trained networks. According to [26] and [27] two types of fine tuning, Shallow fine-tuning (SFT) and deep fine-tuning (DFT). Shallow Fine-tuning works like transfer learning, because earlier layers contain more generic features (e.g: edge, simple color, curves), which are useful for many tasks. So SFT delivers better concentration on the last layers. Contrary to SFT, DFT technique permits training the whole model including earlier layers. For it has needed more computational cost and it increases the risk of overfitting. It can be beneficial approach when the target data is different from pre trained data. For Example, Image of the natural beauty from ImageNet dataset belongs to an actual different area comparative to blood smear images. We used Fine-tuning technique in such a way, we trained the whole network but fully-connected layers were freezed because FC layers were responsible for classification, so we will use the same trick for Fully connected (FC) layers as we did for Fully connected (FC) layers in transfer learning.

IV. RESULT AND DISCUSSIONS

Two types of experiment methods are implemented for the estimation of the proposed technique.

- Experiment result using ALL_IDB dataset
- Experiment result using C-NMC dataset

A. Experimental Setting

Firstly, we converted the size of each image into 224 x 224 resolution. We selected the learning rate 0.001, if the monitored value does not improve then reduced the learning rate by factor point 5. We selected the Adamax optimizer that is an extension of Adam optimizer and the network is trained for 30 epochs for ALL_IDB dataset and 40 epochs for C-NMC dataset. The batch size is set as 30 and 40 respectively in the training process. To avoid overfitting, we applied L2 regularization on the kernel regularizer and L1 regularization on the bias regularizer to regularize the weight of each layer. The L2 value is set 0.016 for kernel regularizer and L1 value is set 0.006 for bias regularizer. We selected a Categorical cross-entropy loss function that calculates the loss. The patience is set to 1 to adjust the learning rate if the monitored value does not improve. The Stop patience value is set to 3 to terminate the training process if the monitored value does not improve continuously. All experiments are performed in google colab, a freely available tool for advanced machine learning. Table 5. are presented with each model parameters that were used in the experiment.

TABLE 5. PARAMETERS OF EACH CLASSIFIER

Approach	Classifier	Total-Parameters	Trainable-Parameters	Non-trainable parameters
Transfer Learning	VGG-16	14,848,578	132,866	14,715,712
	Inception-V3	22,336,034	529,154	21,806,880
	ResNet50	24,120,962	529,154	23,591,808
Fine-Tuning	VGG-16	14,848,578	14,847,554	1,024
	Inception-V3	22,336,034	22,297,506	38,528
	ResNet50	24,120,962	24,063,746	57,216

B. Evaluation Metrics

The performance of the proposed research for the detection of ALL is evaluated using multiple parameters like accuracy, precision, sensitivity and the specificity.

Accuracy gives a percentage of the number of correct predictions predicted by the classifier model overall total predictions made. The formula of accuracy is given below:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad 1$$

Sensitivity is referred to as True Positive Rate (TPR).

Formula of sensitivity is defined as:

$$Sensitivity = \frac{TP}{TP + FN} \quad 2$$

Specificity is also known as True Negative Rate (TNR) and referred a

$$Specificity = \frac{TN}{TN + FP} \quad 3$$

Precision is also called Positive predicted value (PPV) and formula of precision is:

$$Precision = \frac{TP}{TP + FP} \quad 4$$

Whereas; TP=True Positive, TN=True Negative, FP=False Positive, FN=False Negative.

C. Experiment result using ALL_IDB Dataset

In the experiment, we used 3000 images for the classification of blasts and normal cells which were distributed into 80% training and 10% for authenticate and 10% for testing. We selected the Adamax optimizer and the network is trained for 30 epochs for ALL_IDB dataset. The array size is set as 30 in the training activity. We applied transfer learning and fine-tuning techniques on each classifier to get results.

In “Fig. 5”, training loss gradually decreases and after every epoch close to validation loss. We can clearly see no overfitting in “Fig. 5”, and Table 6. represents the evaluation metrics of each classifier that were used in this experiment. We can see clearly in Table 6. The Inception-V3 model gave poor performance based on the transfer learning approach and achieved only 87% accuracy which is very low compared to ResNet50 and VGG16. ResNet50 contains higher accuracy than other VGG16 classifiers.

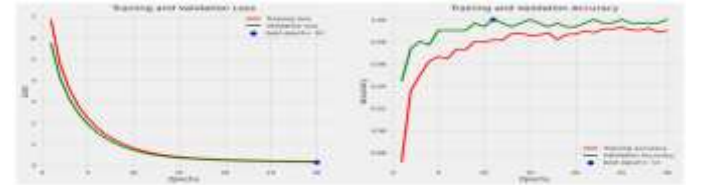


Fig. 5. Accuracy & loss of proposed ResNet-50

TABLE 6. EVALUATION WITHOUT FINE TUNING (BEST VALUES IN BOLD)

Methods	Sensitivity%	Specificity	Precision%	F1-Score%	Accuracy%
VGG-16	99.34	100	100	99	99.67
Inception-V3	94	80.26	82.66	87.11	87.17
ResNet-50	100	100	100	100	100

In “Fig. 6”, training and validation accuracy sharply increase and both are found together at the optimal point in the later stage. At the initial phases of the training segment there was a substantial loss which diminished substantially in the later phases of the training. The vital reason for that loud rise and decline in “Fig. 6”, is the quantity of data for each class in the dataset.

Table 7. represents the evaluation metrics of each classifier that were used in this experiment. We can see clearly in Table 7. Inception-V3 model gave better performance using transfer learning based fine-tuning approach and achieved 100% accuracy. Vgg16 decreased the performance using fine-tuning and achieved 94 percent accuracy which was 6 percent lower than compared to transfer learning. ResNet50 once again achieved 100% accuracy.

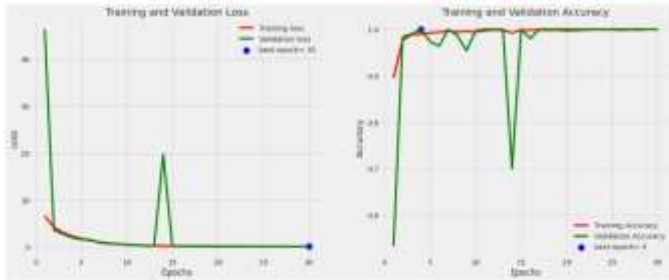


Fig. 6. Accuracy & loss of proposed ResNet-50

TABLE 7. EVALUATION USING FINE TUNING (BEST VALUES IN BOLD)

Methods	Sensitivity%	Specificity	Precision%	F1-Score%	Accuracy%
VGG-16	96.71	92.11	92.45	94.40	94.41
Inception-V3	100	100	100	100	100
ResNet-50	100	100	100	100	100

D. Experiment Result using C-NMC Dataset

In the experiment, we used 6100 images for the training purpose and 533 images for validation. For overall accuracy we used testing data that consisted of 534 original images that were not used in the training phase. We selected the Adamax optimizer and the network is trained for 40 epochs for the C-NMC dataset. The batch size is set as 40 in the training process. We applied transfer learning and Fine tuning technique on each classifier to get results.

In “Fig. 7”, Training and validation loss gradually decrease, from which improves the performance of the model. Table 8. represents the evaluation metrics of each classifier that were used in this experiment. We can see clearly in Table 8.

InceptionV3 achieved almost 81% accuracy. VGG-16 achieved 82.40% accuracy which is higher than InceptionV3. ResNet50 achieved 85.58% accuracy based on transfer learning which is higher than both VGG16 and InceptionV3 models.

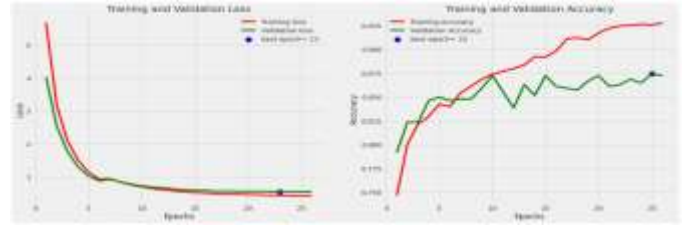


Fig. 7. Accuracy & loss of proposed ResNet-50

TABLE 8. EVALUATION WITHOUT FINE TUNING (BEST VALUES IN BOLD)

Methods	Accuracy%	Sensitivity%	Specificity%	Precision%	F1-Score%
VGG-16	82.40	85.16	76.47	88.57	82.57
Inception-V3	80.90	83.24	75.88	88	81.16
ResNet-50	85.58	88.19	80	90.42	85.68

In the initial stage of training, the sharp rise and decline in “Fig. 8”, is due to quantity the of data for each class in the dataset. while in each cycle, the network inspects all the Images examined, due to that, these ups and downs were reduced as the training procedure advanced. Training had been terminated at epoch 30 after 3 adjustments of learning rate with no improvement. Table 9. represents the evaluation metrics of each classifier. In Table 9. VGG16 achieved 90% accuracy based on a fine-tuning approach which is 8% higher than the transfer learning approach. InceptionV3 achieved 93% accuracy based on fine-tuning which is 12% higher than the transfer learning approach. ResNet50 achieved 97% accuracy which is higher than both of vgg16 and InceptionV3 based on fine-tuning and also 12% higher than transfer learning based ResNet50 model.

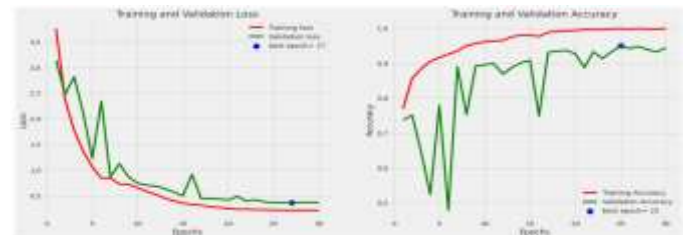


Fig. 8. Accuracy & loss of proposed ResNet-50 Model Based Fine-tuning

TABLE 9. EVALUATION USING FINE TUNING (BEST VALUES IN BOLD)

Methods	Accuracy%	Sensitivity%	Specificity%	Precision%	F1-Score%
VGG-16	90.45	94.23	82.35	91.96	90.38
Inception-V3	93.82	96.43	88.24	94.61	93.78
ResNet-50	97.19	98.63	94.12	97.29	97

V. CONCLUSION AND FUTURE DIRECTIONS

We proposed research which predicts Acute Lymphoblastic leukemia using pre-trained models based on transfer learning and fine-tuning. We used ALL-IDB and C-NMC datasets for

this purpose that consist of healthy and unhealthy images. After experiments we found that a fine-tuning approach is more suitable when the target data is different from pre trained data. For prediction, firstly we preprocessed MBSI images then Augmentation technique implemented to increase the images in the dataset. Three state-of-the-art classifiers ResNet50, InceptionV3 and VGG16 were applied to get best accuracy. ResNet50 achieved 100% accuracy both times based on transfer learning and fine-tuning techniques using ALL_IDB dataset and achieved 97% accuracy using C-NMC dataset. InceptionV3 gave better performance when fine-tuning techniques were applied to it. VGG-16 gave better accuracy based on transfer learning but less performance than ResNet50 and InceptionV3 based on fine-tuning. We will consider different advanced machine learning networks and databases with more samples in future research.

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